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JEE (MAIN+ADVANCED)

Physics

ENTHUSIAST COURSE



STUDY MATERIAL

EM Waves & Wave Optics

English Medium

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ELECTROMAGNETIC WAVES & WAVE OPTICS

KEY CONCEPT

ELECTROMAGNETIC WAVES

Ampere Law :

The general form of Ampere's law (sometimes called the **Ampere-Maxwell law**) as

$$\oint \vec{B}\cdot d\vec{\ell} = \mu_0 (I + I_0) = \mu_0 \left(I + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

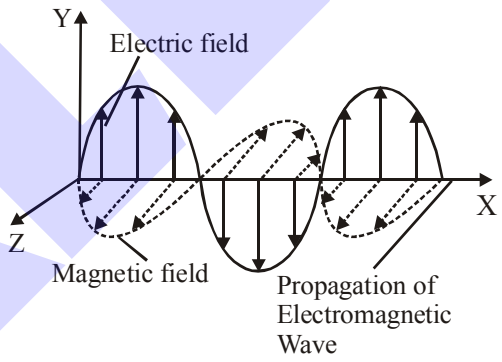
Hence, the displacement current through any surface is given by

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

By considering surface S_2, we can identify the displacement current as the source ofthe magnetic field

ELECTROMAGNETIC WAVES:

In electromagnetic waves, both the field vectors ($\vec{E}$ and $\vec{B}$) vary with time and space and have the same frequency and same phase. In figure, the electric field vector $(\vec{E})$ and magnetic field vector $(\vec{B})$ are vibrating along Y and Z directions and propagation of electromagnetic wave is shown in X-direction.



According to Maxwell the electromagnetic waves are of transverse in nature and they can pass through vacuum with the speed of light (= 3×10^8 ms^{-1}).

The velocity of eletromagnetic wave in a medium is given by

$$v = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$$

where, μ_0, μ_r = absolute permeability of space and relative permeability of medium,

ϵ_0, ϵ_r = absolute permittivity of space and relative permittivity of medium

The velocity of electromagnetic waves of different frequency in vacuum is same but in a medium is different. It is more for red light and less for violet light.

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The energy is shared equally between electric field vector and magnetic field vector.

$$U_{av} = \frac{1}{2}\epsilon_0 E_0^2 = \frac{1}{2}\frac{B_0^2}{\mu_0}$$

It has been found that the velocity (c) of electromagnetic wave in free space is equal to the ratio of amplitude of electric field vector (E_0) and magnetic field vector (B_0) i.e $c = E_0/B_0$.
It was found that the accelerated charge or oscillating charge is a source of electromagnetic waves.

If the plane of electric field is oriented horizontally in respect to the earth, the electromagnetic wave is said to be horizontally polarised. On the other hand, if the plane of electric field vector is oriented vertically the electromagnetic wave is said to be vertically polarised.

The polarisation of electromagnetic wave is mainly the function of the antenna orientation.

Light may be polarized by passing it through a sheet of commercial material called Polaroid

Malus' Law
Suppose we have a second piece of Polaroid whose transmission axis makes an angle θ with that of the first one.

If $I_0 \cong E^2$ is the intensity between the two Polaroids, the intensity transmitted by both of them would be: $I(\theta) = I_0 \cos^2\theta$.

Brewster's law

$$\tan\theta_P = \frac{n_2}{n_1} : \textbf{Brewster's law}$$

θ_P is the angle of incidence of unpolarized light which makes the reflected light completely polarized in the perpendicular direction to the plane of incidence (Sir David Brewster, 1812). When the angle of incidence of the initially unpolarized light is θ_P, the reflected and refracted rays are perpendicular to each other.

INTERFERENCE OF LIGHT
When two light waves of same frequency with zero initial phase difference or constant phase difference superimpose over each other, then the resultant intensity in the region of superimposition is different from the sum of intensity of individual waves.
This modification in intensity in the region of superposition is called interference.

(a) Constructive interference
When resultant intensity is greater than the sum of two individual wave intensities [$I > (I_1 + I_2)$], then the interference is said to be constructive.

(b) Destructive interference
When the resultant intensity is less than the sum of two individual wave intensities [$I < (I_1 + I_2)$], then the interference is said to destructive. There is no violation of the law of conservation of energy in interference. Here, the energy from the points of minimum energy is shifted to the points of maximum energy.

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TYPES OF SOURCES

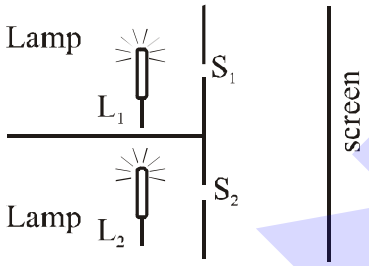
(a) Coherent sources

Two sources are said to be coherent if they emit light waves of the same wave length and start with same phase or have a constant phase difference.

Note : Laser is a source of monochromatic light waves of high degree of coherence.

(b) Incoherent sources

Two independent monochromatic sources, emit waves of same wavelength. But the waves are not in phase. So they are incoherent. This is because, atoms cannot emit light waves in same phase and these sources are said to be incoherent sources.



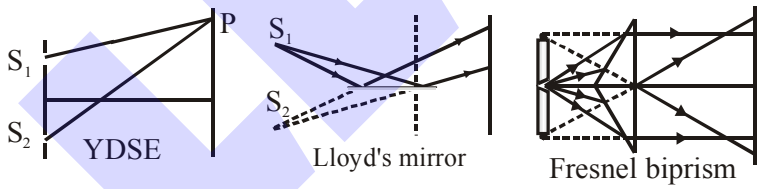
independent source
(noncoherent source)

METHOD FOR OBTAINING COHERENT SOURCES

(a) Division of wavefront

In this method, the wavefront is divided into two or more parts by use of mirrors, lenses or prisms.

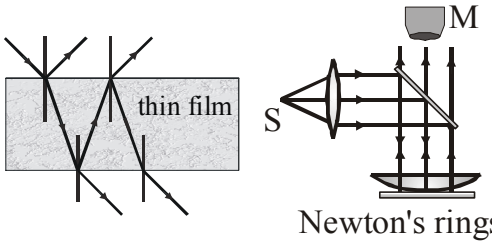
Example : Young's double slit experiment. Fresnel's Biprism and Lloyd's single mirror method.



(b) Division of amplitude

The amplitude of incoming beam is divided into two or more parts by partial reflection or refraction. These divided parts travel different paths and are finally brought together to produce interference.

Example : The brilliant colour seen in a thin film of transparent material like soap film, oil film, Michelson's Interferro Meter, Newtons' ring etc.



Newton's rings

If two coherent waves with intensity I_1 and I_2 are superimposed with a phase difference of ϕ, the resulting wave intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \cos\phi$$

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The phase difference between two waves at a point will depend upon

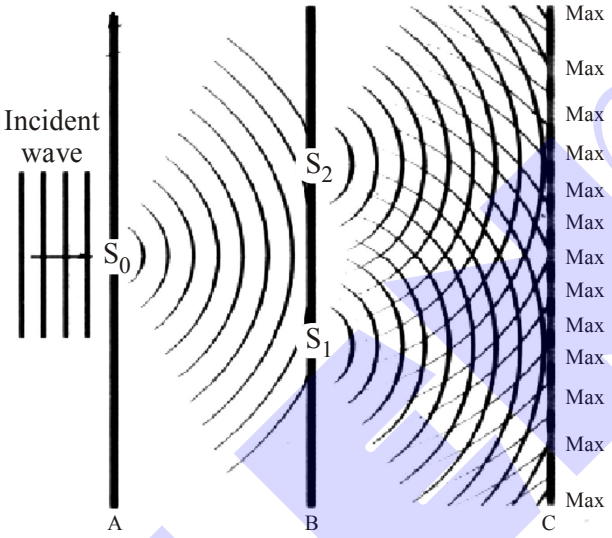
(i) the difference in path lengths of two waves from their respective sources.(geometrical path difference)

(ii) the refractive index of the medium (media)

(iii) phase difference at source (if any).

(iv) In case, the waves suffer reflection, the reflected wave differs in phase by π with respect to the incident wave if the incidence occurs in rarer medium. There would be no phase difference if incidence occurs in denser medium.

YOUNG'S DOUBLE SLIT EXPERIMENT

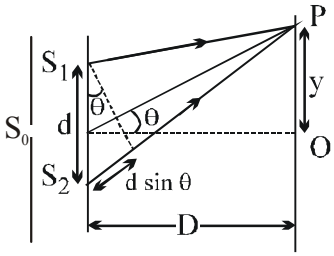


(i) If $d << D$

$\Delta x = S_2P - S_1P = d\sin\theta$

If $\lambda << d$ then $\sin\theta \approx \theta \approx \tan\theta$ as when P is close to D so θ is small.

$$\Delta x = d\left(\frac{y}{D}\right) = \frac{yd}{D}$$



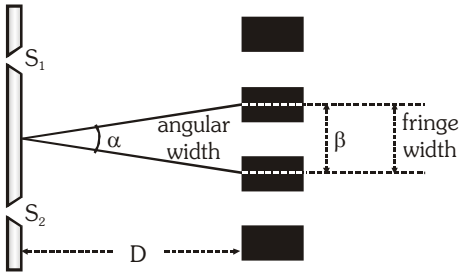
(ii) For maxima $\frac{yd}{D} = n\lambda$

or $y = 0, \pm\frac{D\lambda}{d}, \pm\frac{2D\lambda}{d}$

(iii) For minima $\frac{yd}{D} = [n + (1/2)]\lambda$ or $y = \pm\frac{D\lambda}{2d}, \pm\frac{3D\lambda}{2d}, \pm\frac{5D\lambda}{2d}$, so on

(iv) Fringe width = the distance between two successive maximas or minimas $\beta = \frac{\lambda D}{d}$

(v) **Angular Fringe width**

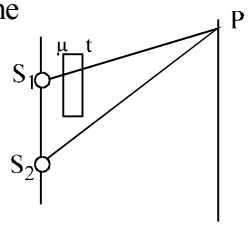


Angular Fringe Width $\alpha = \frac{\beta}{D}$, $\alpha = \frac{\lambda}{d}$ $\left[\because \frac{\beta}{D} = \frac{\lambda}{d}\right]$

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DISPLACEMENT OF FRINGE PATTERN

When a film of thickness t and refractive index μ is introduced in the path of one of the source's of light, then fringe shift occurs as the optical path difference changes.



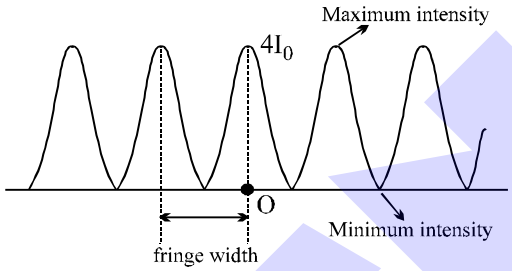
Optical path difference at P is

$\Delta x = S_2P - [S_1P + \mu t - t] = S_2P - S_1P - (\mu - 1)\, t = y.\,(d/D) - (\mu - 1)t$

$\Rightarrow$ The fringe shift is given by $\Delta y = \dfrac{D(\mu - 1)t}{d}$

INTENSITY VARIATION ON SCREEN

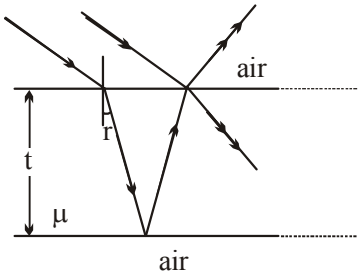
If I_0 is the intensity of light beam coming from each slit, the resultant intensity at a point where they have a phase difference of ϕ is $I = 4I_0 \cos^2 \dfrac{\phi}{2}$, where $\phi = \dfrac{2\pi(d \sin\theta)}{\lambda}$



INTERFERENCE AT THIN FILM

optical path difference $= 2\mu t \cos r$

$= 2\mu t$ (in case of near normal incidence)



(a) For interference in reflected light

(i) Condition of minima $2\mu t \cos r = n\lambda$

(ii) Condition of maxima $2\mu t \cos r = \left(n + \dfrac{1}{2}\right)\lambda$

(b) For interference in reflected light

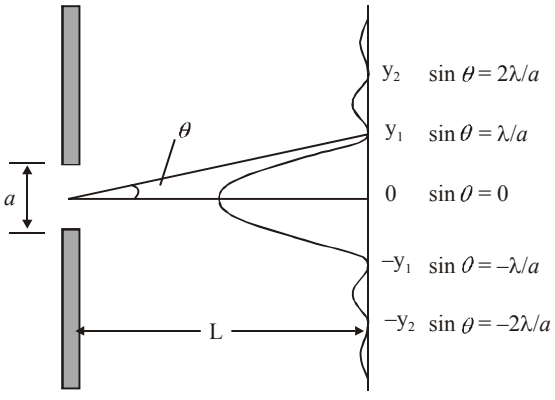
(i) Condition of maxima $2\mu t \cos r = n\lambda$

(ii) Condition of minima $2\mu t \cos r = \left(n + \dfrac{1}{2}\right)\lambda$

DIFFRACTION

When light passes through a narrow slit of width comparable to the wavelength of light, the light flares out of the slit; this bending or spreading of waves is called diffraction.

Properties of dark & bright fringes:

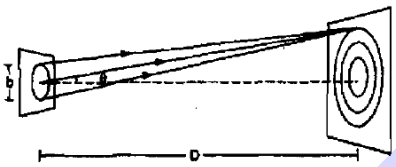


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(a) It results from superposition of secondary wavelets originating from various parts of single coherent source.
(b) Diffraction fringes are never of equal width.
(c) Intensity of all the bright fringes is not the same.
(d) Dark fringes are not perfectly dark.

FRAUNHOFER DIFFRACTION BY A CIRCULAR APERTURE

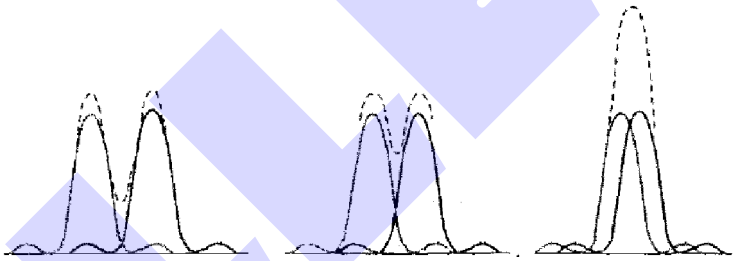
The mathematical analysis shows that the first dark ring is formed by the light diffracted from the hole at an angle θ with the axis $\sin\theta = \frac{1.22\lambda}{b}$.



The radius of the diffraction disc is given by $R = \frac{1.22\lambda D}{b}$,

LIMIT OF RESOLUTION

The fact that a lens forms a disc image of a point source, puts a limit on resolving two neighboring points imaged by a lens.



well resolved just resolved unresolved.

For two objects to be barely resolved, the angular separation between them should be at least:

$$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{b}\right)$$

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